Benchmark Test: GEV Hazard Curve Model (MKM-GH-001)

Model	MKM-GH-001 GEV Hazard Curve Model
Generated	2026-10-19
Code under test	src/models/hazard/gev.py sha256 cca7d20506171a85... commit 8bbb73347e54
Numerics	numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1
Wolfram	not run
Data	synthetic samples from known parameters, seed 20261019

Findings

Check	Result	Outcome
Closed-form return levels and exceedance, T <= 10,000 (tolerance 1e-09)	worst level 3.0e-11 (below_gumbel_switch/T=10000); worst exceedance 4.2e-10 (below_gumbel_switch/T=10000)	PASS
Wolfram confirms the reference	Not run. The closed forms rest on one reference implementation (mpmath) until it is.	PENDING
Maximum-likelihood fit reaches the maximum	largest log-likelihood gain found by the independent fit: 2.1e-06	PASS
Sampling uncertainty in the 1-in-1000 level (synthetic, known truth)	fitted / true level ranges 1.02x to 4.46x across samples	FINDING

Sign convention: the 1-in-1000 level for xi = 0.2 is 398.07; passed to scipy unflipped it would be 174.88. GEVFitter flips it correctly.

Return levels and exceedance

Case	T	Reference level	Level err	P(X > x_T)	Exceed. err
gumbel	10	145.0073465462489057	1.3e-16	1.00e-01	6.4e-16
gumbel	100	192.0029845355315999	2.3e-17	1.00e-02	6.7e-16
gumbel	200	205.9162428507005173	2.1e-17	5.00e-03	6.8e-16
gumbel	1,000	238.1451014104743299	2.0e-17	1.00e-03	6.4e-16
gumbel	10,000	284.2058073978544878	7.7e-15	1.00e-04	1.1e-13
gumbel	1,000,000	376.3102011592813154	1.5e-12	1.00e-06	2.9e-11
frechet_moderate	10	156.8427406502533736	1.2e-17	1.00e-01	1.6e-16
frechet_moderate	100	250.9365281717156733	2.0e-16	1.00e-02	1.0e-15
frechet_moderate	200	288.3954459358693725	3.9e-17	5.00e-03	6.9e-16
frechet_moderate	1,000	398.0673452308130602	7.2e-17	1.00e-03	5.4e-16

Case	T	Reference level	Level err	P(X > x_T)	Exceed. err
frechet_moderate	10,000	630.9510346753842052	2.2e-14	1.00e-04	1.1e-13
frechet_moderate	1,000,000	1584.893033971736286	5.8e-12	1.00e-06	2.9e-11
frechet_heavy	10	183.2313049904440671	3.3e-17	1.00e-01	1.3e-16
frechet_heavy	100	458.9970676051084418	5.0e-16	1.00e-02	8.8e-16
frechet_heavy	200	624.9772841994358159	2.8e-16	5.00e-03	1.1e-15
frechet_heavy	1,000	1324.594744018793456	2.6e-16	1.00e-03	1.1e-15
frechet_heavy	10,000	4059.899997083177073	5.5e-14	1.00e-04	1.1e-13
frechet_heavy	1,000,000	40059.98999999708333	1.4e-11	1.00e-06	2.9e-11
weibull_bounded	10	136.2418690304628677	8.7e-17	1.00e-01	4.5e-16
weibull_bounded	100	160.1492852680379470	2.6e-17	1.00e-02	3.6e-16
weibull_bounded	200	165.3253886601811830	4.4e-17	5.00e-03	1.7e-16
weibull_bounded	1,000	174.8786226255216708	5.5e-18	1.00e-03	1.1e-15
weibull_bounded	10,000	184.1509095786729828	2.0e-15	1.00e-04	1.1e-13
weibull_bounded	1,000,000	193.6904259242404243	1.9e-13	1.00e-06	2.9e-11
near_zero_shape	10	145.0073465968904368	4.2e-17	1.00e-01	6.6e-17
near_zero_shape	100	192.0029847471453293	5.1e-17	1.00e-02	4.0e-16
near_zero_shape	200	205.9162431311567803	9.1e-17	5.00e-03	1.4e-15
near_zero_shape	1,000	238.1451018875760571	1.6e-17	1.00e-03	1.1e-15
near_zero_shape	10,000	284.2058082461489774	7.6e-15	1.00e-04	1.1e-13
near_zero_shape	1,000,000	376.3102030679645058	1.5e-12	1.00e-06	2.9e-11
below_gumbel_switch	10	145.0073465467553210	3.5e-12	1.00e-01	2.4e-11
below_gumbel_switch	100	192.0029845376477372	1.1e-11	1.00e-02	1.1e-10
below_gumbel_switch	200	205.9162428535050800	1.4e-11	5.00e-03	1.4e-10
below_gumbel_switch	1,000	238.1451014152453472	2.0e-11	1.00e-03	2.4e-10
below_gumbel_switch	10,000	284.2058074063374327	3.0e-11	1.00e-04	4.2e-10
below_gumbel_switch	1,000,000	376.3102011783681472	5.2e-11	1.00e-06	9.7e-10

Maximum-likelihood fit

Series	Fit	n	xi model	xi indep.	xi true	LL gain	x_100 model	x_100 true	x_1000 model	x_1000 true
n30_xi0.1	gev	30	0.2449	0.2449	0.10	1.1e-07	6.182	5.336	9.741	6.981
n30_xi0.1	gumbel	30	0.0000	0.0000	0.10	-4.7e-15	5.028	5.336	6.011	6.981

Series	Fit	n	xi model	xi indep.	xi true	LL gain	x_100 model	x_100 true	x_1000 model	x_1000 true
n50_xi0.3	gev	50	0.6588	0.6588	0.30	9.0e-08	13.774	6.967	54.618	12.256
n50_xi0.3	gumbel	50	0.0000	0.0000	0.30	-1.8e-15	5.856	6.967	7.262	12.256
n200_xi-0.1	gev	200	-0.0832	-0.0833	-0.10	2.1e-06	4.520	4.475	5.087	4.995
n200_xi-0.1	gumbel	200	0.0000	0.0000	-0.10	-5.9e-15	4.767	4.475	5.659	4.995
n30_xi0.4	gev	30	0.7073	0.7073	0.40	1.0e-07	16.839	8.297	76.048	17.846
n30_xi0.4	gumbel	30	0.0000	0.0000	0.40	-2.9e-15	7.065	8.297	8.996	17.846

Limitations

• This is a benchmark of the implementation, not independent validation: it shows the model computes what its formulae say, not that the formulae suit the data.

• The fit section uses seeded synthetic samples unless a gauge export was supplied; see 'Data' above.

• GEVFitter fits the GEV to every storm's peak level. The GEV is the limiting law for block maxima (e.g. annual maxima); the per-storm choice is a methodology question this benchmark does not answer.

• Parameter uncertainty is the dominant tail risk at the sample sizes tested; return levels carry no confidence interval in the model today.